

Local Labor Supply Elasticities in Mexico

Lorenzo Aldeco Leo*

Jorge Pérez Pérez*

Abstract

We estimate the inverse elasticity of labor supply at the local-labor-market level in Mexico for 1990-2020, using shift-share instruments to address the endogeneity of local labor demand. Excluding agriculture, a 1% demand-driven increase in male employment raises male wages by 0.36%, a more elastic response than in the U.S. When including agriculture, however, male local labor supply appears inelastic, consistent with demand shocks reallocating employees out of agricultural work. We find evidence that female local labor supply is even more elastic: demand shocks raise female employment, particularly among married women, with no detectable effect on wages. Our results suggest local demand shocks in Mexico mainly create jobs rather than raise wages.

Keywords: Labor supply; Local labor markets; Mexico; Instrumental variables

JEL Codes: J22, J23, C23, C26, O54

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1 Introduction

How local labor markets respond to shifts in labor demand is central to understanding who gains from local economic growth. The local labor supply elasticity that governs this response is a key parameter in the estimation of the effects of local demand shocks and the evaluation of place-based policies (Bartik, 1991; Moretti, 2011; Kline and Moretti, 2014; Neumark and Simpson, 2015; Notowidigdo, 2020). When local labor supply is inelastic, a positive demand shock raises the wages of already employed workers more than it creates new employment; when it is elastic, the shock mostly draws new workers into employment, with more muted effects on average worker wages. Despite the importance of measuring the local labor supply elasticity for developing-country labor markets, high-quality estimates of it come mostly from the U.S. (Adão, Kolesár, and Morales, 2019; Goldsmith-Pinkham, Sorkin, and Swift, 2020). In this paper, we estimate the inverse local labor supply elasticity for men and women in Mexico over 1990-2020. This parameter measures the percent wage increase required to draw one percent more workers into employment in the local labor market. Mexico is a particularly attractive setting to study these responses for several reasons. Over the last four decades, Mexico’s local labor markets have been exposed to several large shocks originating abroad, including the North American Free Trade Agreement (NAFTA), China’s entry into the World Trade Organization (WTO), the Global Financial Crisis, and the U.S.-China trade war. These shocks both provide variation in local labor demand and help us better understand how a middle-income open economy responds to global shocks. Furthermore, Mexico’s labor market has the markers and challenges of a developing labor market: low and slow-growing average wages, low female labor force participation, and persistent geographical inequality (Aldeco, Osuna, and Pérez Pérez, 2025). The setting then allows us to speak to the role that labor supply and demand can have in addressing some of these challenges.

While relevant for policy, the estimation of local labor supply elasticities in developing countries is subject to some challenges. Although this estimation has a long history within labor economics –starting with Bartik (1991)–, only recent advances in the econometrics of shift-share designs have clarified the assumptions under which the instruments most commonly used in these empirical strategies identify causal effects, as well as how to conduct valid inference with them (Adão, Kolesár, and Morales, 2019; Goldsmith-Pinkham, Sorkin, and Swift, 2020; Borusyak, Hull, and Jaravel, 2022; Borusyak, Hull, and Jaravel, 2025). We follow this recent literature and instrument local employment growth with gender-specific shift-share measures of local labor demand shocks, relying on many as-good-as-random industry shocks for identification and exposure-robust inference (Borusyak, Hull, and Jaravel, 2022).

This approach allows exposure shares to be correlated with local supply shifters, which is attractive in our setting, given that some important factors that influence labor supply, such as conditional cash transfers (Parker and Todd, 2017) and migration networks (Munshi, 2003), are unobserved in our data but distributed unevenly across local labor markets.

A related challenge is the definition of local labor markets themselves. Disaggregated labor market data is usually reported at the level of political units, which may not correspond to the scale at which labor market shocks appear, complicating estimation. To have estimates that are comparable across countries, local labor market definitions must also be constructed consistently. We address this issue by using harmonized Census microdata and local labor market definitions built from commuting flows with the same method used to construct U.S. commuting zones (Aldeco et al., 2024), making our estimates comparable to those in the U.S. literature.

Our estimates correspond to a Marshallian labor supply elasticity: they incorporate both the substitution and income effects that follow a labor demand shock, and capture only the extensive margin of employment. They are also local, so they include changes in migration, commuting, and sectoral relocation within the local labor market. These features make them comparable to those used in the evaluation of place-based policies, but lead them to differ from individual extensive- and intensive-margin labor supply elasticities, as well as from estimates of the intertemporal elasticity of individual labor supply (Chetty et al., 2011).

We use data from the Mexican Population Censuses of 1990, 2000, 2010, and 2020, collected by INEGI and harmonized by Banco de México to make them comparable across years. We leverage the local labor market definitions in Aldeco et al. (2024), which partition Mexico on the basis of municipality-level commuting flows. Our main variables are the 10-year changes in log total employment and log average wages in each local labor market, which we measure separately for men and women.

We find evidence that non-agricultural local labor supply in Mexico is elastic for men and highly elastic for women. In our preferred specification, which excludes agricultural employment, an increase of 1% in male employment driven by local demand shocks is accompanied by a 0.36% increase in male wages. This response is more elastic than estimates for the U.S., of about 0.8 (Adão, Kolesár, and Morales, 2019). Including agriculture yields an inelastic male local labor supply, with a point estimate of 1.35, consistent with non-agricultural local demand shocks mostly drawing workers from lower-paid agricultural employment, rather than increasing local labor force participation. Our estimates point to female local labor supply being highly elastic: first-stage estimates of the employment response to a local labor

demand shock are consistently positive and statistically significant, but our preferred estimate of wage responses is low and statistically insignificant. Among women, employment responses to demand shocks are larger for married women, consistent with them being an important pool of marginal workers available to local employers when responding to these shocks.

These results contribute to explaining why real wage distributions across Mexican local labor markets have remained stable since the 1990s despite the large external shocks the country has faced (Aldeco et al., 2024), and why trade shocks in Mexico usually induce larger employment than wage responses, often with stronger effects on female than male employment (Trachtenberg, 2019; Chen, Novy, and Solórzano, 2025; Estefan, 2026). Our estimates suggest that external demand shocks to Mexican local labor markets create jobs to a larger extent than they raise the wages of already-employed workers, and that drawing women into employment is an important margin through which local labor markets accommodate these shocks. This suggests that demand-side policies in the Mexican labor market are most apt to increase employment levels, but that their role in increasing aggregate wages may be limited. In this light, our results highlight that increased aggregate wages are more likely to be achieved efficiently through supply-side policies, such as improving human capital.

The rest of the paper is structured as follows. We review the literature and state our contributions in Section 2. Section 3 describes our data. We describe our empirical strategy and results in sections 4 and 5, respectively. Section 6 concludes.

2 Literature Review

We contribute mainly to three strands of literature. First, we add to the set of works that estimate employment elasticities at the local-labor-market level. In the U.S., a strand of literature has focused on isolating the effects of local labor demand shocks, following the seminal contribution of Blanchard and Katz (1992), who separate them using a structural VAR model. Bartik (1991) provided the first estimate of the local labor supply elasticity using shift-share demand shocks. Amior and Manning (2018) argue that labor demand shocks are persistent at the local level. Monte, Redding, and Rossi-Hansberg (2018) show that local employment elasticities to demand shocks depend on the degree of connectedness of a local labor market to surrounding markets. More recently, in the context of analyzing the assumptions and properties of empirical designs based on shift-share “Bartik” instruments, Adão, Kolesár, and Morales (2019) and Goldsmith-Pinkham, Sorkin, and Swift (2020) provide estimates of

the local labor supply elasticity for the United States. We bring these estimation techniques, along with the insights from Borusyak, Hull, and Jaravel (2022), to the estimation of local labor supply elasticities in a developing country setting. By virtue of the local labor market definitions we use, which follow the construction used to build U.S. commuting zones, our results are closely comparable to those in this literature. We find evidence that male local labor supply is more elastic than that found for the U.S., with evidence of low average female wage responses.

Second, we add to a rich literature that studies the effects of demand shocks on the Mexican labor market. The majority of these papers use variation in local exposure to trade shocks and focus on manufacturing as opposed to aggregate local employment. They consistently find stronger effects on employment than on wages, indicating that local labor supply is elastic for specific sectors at the local level. Blyde et al. (2023) estimate the effect of the “China shock” (Autor, Dorn, and Hanson, 2013) using their own definition of Mexican commuting zones, and find employment responses that are much larger and more persistent than wage ones. Using the same commuting-zone definitions, Robertson, Halliday, and Vasireddy (2020) focus on responses in the textile and apparel sector to changes in the Multi Fibre Agreement, which freed textile imports into the U.S., and also find employment effects larger than wage ones at the metropolitan-area level. Estefan (2026) finds that variations in U.S. demand for Mexican *maquiladora* exports are reflected in changes in employment and local labor force participation at the state level, with no measurable wage responses. At the firm level, Trachtenberg (2019) finds positive employment effects and null wage effects for Mexican manufacturing firms exposed to export demand from the North American Free Trade Agreement (NAFTA). Similarly, Chen, Novy, and Solórzano (2025) show employment responded more strongly than wages (conditional on worker characteristics) among exporting firms in response to the China-U.S. trade war. Notably, these last two papers analyze effects by gender, and find that women’s employment responses are stronger than men’s, suggesting that female employment may be more elastic than male employment.

We contribute to this literature by shifting attention away from manufacturing and towards the aggregate effects of demand shocks on employment and wages at the local level. We focus on estimating the local labor supply elasticity, which governs the effects of a generic shift in labor demand and so is likely to be of specific interest to policymakers.^{1,2} While the results

¹Beyond trade shocks, demand shocks whose effects depend on this elasticity also include local fiscal stimuli and place-based policies.

²Chiquiar, Covarrubias, and Salcedo (2017) and Vazquez and Winkler (2026) estimate the effects of trade shocks on local employment and wages including all employment sectors, but do not target the local labor

from the literature on trade shocks are informative about local responses in the manufacturing sector, their focus on that sector may paint only a partial picture about the total effects of local shocks. Demand shocks can induce sectoral reallocation within a local labor market, or spill over into labor demand for other sectors, making sectoral and total local responses different. For instance, a manufacturing-sector expansion that draws workers from lower-wage, non-manufacturing employment within the same local labor market is consistent with elastic labor supply to manufacturing, while total employment across sectors can remain constant, i.e., total local labor supply can still be inelastic. We directly estimate the aggregate local labor supply, including and excluding agriculture. Consistent with some sectoral reallocation, we find that labor supply at the local level when excluding agriculture is elastic, while estimated supply including all sectors is inelastic.

Last, we contribute to a smaller literature that studies the differences in labor supply responses to demand shocks between men and women in developing-country settings (Gaddis and Pieters, 2017; Chauvin, 2024; Ben Yahmed and Bombarda, 2020). Closest to us is Chauvin (2024), which shows that male and female labor supply respond differently to similar shocks, with males more likely to migrate in response to positive shocks and females more likely to move in and out of the labor force in Brazil. We analyze responses to demand shocks by gender, and bring in the dimension of wages to make statements about male and female local labor supply elasticities. Consistent with Chauvin (2024), we find heterogeneous labor supply responses across men and women, with female supply being more elastic to local shocks. These estimates point to stronger female labor supply responses along the employment margin than the wage margin, providing suggestive evidence that local demand shocks are more likely to increase female employment than wages.

3 Data

Our main data sources are the Mexico Population Censuses and Intercensal Surveys from 1990, 2000, 2010, 2015, and 2020, collected by Mexico’s national statistics office, *Instituto Nacional de Geografía y Estadística* (INEGI). We use datasets harmonized by Banco de México (Aldeco et al., 2024), which edited them to make them comparable across years. The datasets include individual-level information on demographics, education, and labor market outcomes, including employment, wages, and 3-digit NAICS sector identifiers.

Given that many labor market dynamics happen locally (Moretti, 2011), we focus on local supply elasticity.

labor markets (LLMs) as the main unit of analysis. In our application, LLMs are groups of municipalities that are connected by commuting. Aldeco et al. (2024) describe the method for building LLMs: it first uses commuting flows and municipality employment size to construct a measure of commuting proximity between municipalities, which it then uses to cluster municipalities into LLMs. This method is the same as that used to construct U.S. commuting zones (Tolbert and Sizer, 1996), making our results comparable to those in the U.S. setting. According to the definitions in Aldeco et al. (2024), Mexico is partitioned into 777 local labor markets.^{3,4} Each Census and Intercensal Survey is representative at the municipality level, allowing us to construct variables for the whole set of Mexican LLMs for each Census year.

We aim to estimate local labor supply elasticities separately for men and women, so our labor market variables are employment totals and average wages in each LLM by gender. Local labor supply responses among women are heterogeneous in ways that a single elasticity may not well summarize, since participation decisions in other settings depend on marital status and on childcare demands within the household (Kimmel and Kniesner, 1998; Killingsworth and Heckman, 1986; Attanasio et al., 2018; Addati et al., 2018). To account for this heterogeneity in the analysis, we construct sub-samples of women according to marital status and by the presence and age of children in the household. We group married and cohabiting women together and other categories (single, widowed, separated) in another. Similarly, to capture that childcare demands vary with household composition and children’s age, we bin women in households with children younger than 6 years old together, separately from the rest (including those without children). This results in four marriage-status-by-childcare-demand bins.⁵

We restrict the sample to individuals 15 years of age or older, corresponding to the legal working age in Mexico. Employment includes salaried workers and the self-employed. We define a person as employed if they report having worked during the week they are surveyed,

³In Mexico, new municipalities are sometimes formed, taking territory from existing ones. To address this and maintain the geographical consistency of LLMs, Aldeco et al. (2024) assign each municipality-year to the 1990 municipality that contributed the most territory to it, and construct LLMs by clustering 1990-dated municipalities.

⁴While the commuting links between municipalities may change over time, to maintain comparability across years we assume LLM definitions stay constant. Our definitions for local labor markets are constructed using 2010 commuting flows, which sit approximately at the midpoint of the study period.

⁵Childcare is more likely to fall on mothers, but Census data does not allow us to identify family relationships within the household consistently. The Census reports the relationship of each respondent to the head of household, but not between non-head members. We then use the presence of young children in the household as a proxy for time demands from childcare.

following INEGI’s catalog of individuals’ main activity. Our main independent variable is employment at the LLM level. We assign employment by place of residence rather than place of work, consistent with low commuting across LLMs.⁶ We aggregate individual employment status to the LLM level using sampling weights, and our main regressor is the 10-year difference in log employment across Census years.⁷

Our main outcome is the average 10-year difference in the log wage in the local labor market. The Censuses and Intercensal Surveys report monthly labor income from the main job, which we convert to constant 2015 pesos using the national price index (*Indice Nacional de Precios al Consumidor*). To build wage variables, we first restrict the sample to employed individuals reporting positive labor income, excluding the bottom and top 1% of the wage distribution within each LLM-year. We then build LLM-level average wages as logs of weighted means of individual labor income, using survey sampling weights. Our dependent variable is the 10-year difference in this LLM-level log wage, taken between consecutive Census years.

We include a vector of LLM-level controls measured at the start of each 10-year period. These are the age composition of the local population, its schooling composition, and the share of the population living in rural localities, defined by INEGI as localities with fewer than 2,500 inhabitants.⁸ The age and education bin shares also account for changes in the composition of the employed population that would otherwise be attributed to wage growth. We also include time fixed effects as controls to absorb national shocks common to all local labor markets in each 10-year period.

The construction of our instrumental variable requires two additional inputs, both of which come from the source Census microdata. The first is the industrial composition of each LLM, obtained from the harmonized sector variable, which follows the 3-digit NAICS classification. The second input is the set of industry-level employment growth rates, which we compute as national employment growth by industry over each 10-year period. Section 4 describes how these two inputs are combined into a measure of local labor demand shocks.

⁶According to Aldeco et al. (2024), 94.2% of 2010 commuting flows happen within local labor markets.

⁷To keep the year difference between consecutive periods constant and equal to 10, we do not include data from the 2015 Intercensal Survey in the analysis.

⁸We use as age shares the share of the local population younger than 20, between 20 and 40, between 40 and 60, between 60 and 80, between 80 and 100, and older than 100, inclusive of the lower end of each bin. For education, we use less than 6, between 6 and 12, between 12 and 15, between 15 and 18, and over 18, inclusive of the upper end of each bin. These correspond approximately to no basic education, finished basic education, finished secondary education, some tertiary, finished tertiary, and post-graduate education.

4 Empirical Strategy

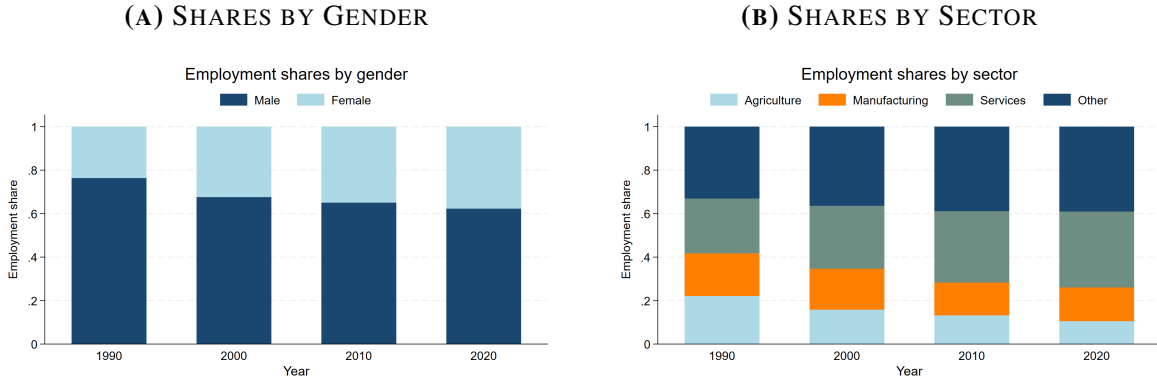
We are interested in the inverse elasticity of local labor supply, β , in the relation

$$\Delta \ln w_{jgt} = \beta \Delta \ln e_{jgt} + \Delta \ln \nu_{jgt}. \quad (1)$$

Here, the time differences Δ are in 10-year intervals, as in our data; j indexes local labor markets (LLMs), t indexes Census years, g denotes gender, w_{jgt} are wages, e_{jgt} are employment levels, and $\Delta \ln \nu_{jgt}$ is a labor supply shock. The parameter β measures the percent wage increase required to draw one percent more workers into employment in the local labor market.

We estimate the elasticity separately for men and women, given the different supply responses that have been documented in the literature (see, e.g., Goldin, 1990; McClelland, Mok, and Pierce, 2014) and to better understand the role of gender differences in local labor markets. As Figure 1 shows, the share of women in total employment grew substantially over the study period, pointing to female labor supply and demand that followed a different pattern from male ones.⁹

FIGURE 1: GENDER AND SECTOR EMPLOYMENT SHARES



Note: This figure shows shares of total employment by gender (Panel A) and by sector (Panel B) in each Census year. Employment includes salaried and self-employed workers aged 15 and older who report having worked during the survey reference week, aggregated using sampling weights. The sectors in Panel B are aggregations of 3-digit NAICS industries; Other includes non-agricultural primary activities, construction, government, and unspecified activities.

Importantly for interpretation, the elasticity we target is Marshallian and on the extensive

⁹Figure A1 shows scatter plots of male and female Bartik shocks, at the LLM and year level. Visually, they show that within an LLM and a year, shocks for both genders varied substantially.

margin. It is Marshallian in the sense that $1/\beta$, the reduced-form labor supply elasticity, incorporates both the substitution and income effects that follow a labor demand shock; and it captures only the extensive margin of employment, i.e., it abstracts from changes in hours worked. Given the nature of our empirical strategy, our estimates are then informative about the local responses to demand shocks and comparable to the local labor market elasticities estimated in the labor and trade literatures and used in the evaluation of place-based policies (Bartik, 1991; Moretti, 2011; Neumark and Simpson, 2015; Adão, Kolesár, and Morales, 2019; Notowidigdo, 2020). They are, however, silent about other concepts of the labor supply elasticity which are of interest to other literatures.¹⁰ Figure 2 illustrates β as the slope of local labor supply, i.e., the response of local labor market wages to a shift in employment driven by a demand shock. The left panel illustrates wages and employment effects when local labor supply is inelastic, which, given our estimation equation, corresponds to $\beta > 1$ and implies that shocks are accommodated more along the wage margin than the employment margin. The right-hand-side panel instead illustrates the case when $\beta < 1$, i.e., the supply is elastic, and demand shocks result in more local employment growth than wage growth.

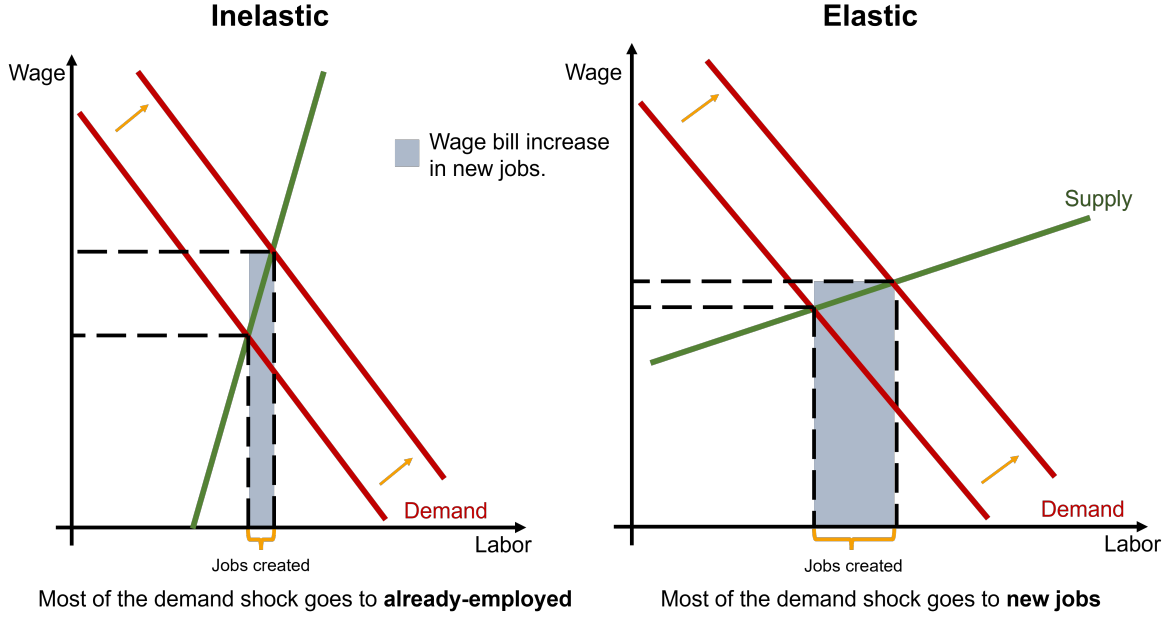
Observed changes in wages and employment are equilibrium outcomes, so the labor supply shocks $\Delta \ln \nu_{jgt}$ are correlated with employment: to the extent that supply and demand are correlated, observed wages will vary for reasons unrelated to the slope of the supply curve. Tracing out the shape of local labor supply then requires variation in wages and employment driven by local labor demand. We add additional structure to the labor supply shocks $\Delta \ln \nu_{jgt}$, modeling them as a function of demographics, national-level shocks, and unobserved local shocks. Our estimating equation is

$$\Delta \ln w_{jgt} = \beta \Delta \ln e_{jgt} + D'_{jgt} \rho + \gamma_t + \varepsilon_{jgt}, \quad (2)$$

where $t = \{2000, 2010, 2020\}$ indexes the end of each 10-year period in our data, D_{jgt} is the vector of start-of-period controls described in Section 3, γ_t are year fixed effects, and ε_{jgt} is an unobserved local shock possibly correlated with $\Delta \ln e_{jgt}$. Estimating (2) by OLS does not recover β , since local demand shocks move wages and employment in the same direction and are absorbed into ε_{jgt} .

¹⁰Among these, the Frisch intertemporal labor supply elasticity that parametrizes supply responses in the New Keynesian Phillips curve; the Hicksian elasticity of labor supply that governs responses to labor taxation; and the upward-sloping labor supply faced by firms with monopsony power.

FIGURE 2: ILLUSTRATION OF THE LOCAL LABOR SUPPLY ELASTICITY



Note: This figure illustrates the effects of a demand shock in a local labor market when local labor supply is inelastic (left panel) and elastic (right panel). Demand is shown in red and supply in green. The shaded area corresponds to the part of the increased wage bill that is paid to jobs created by the demand shock.

We instrument employment growth with a shift-share measure of local labor demand shocks,

$$z_{jgt} = \sum_k \underbrace{s_{kjt0}}_{\text{Shares}} \underbrace{r_{kgt}}_{\text{Shifts}}, \quad (3)$$

where s_{kjt0} is the share of employment in industry k in local labor market j at the start of the study period (1990), and r_{kgt} is the national employment growth rate of industry k for gender g over the period. The shares measure the exposure of each local labor market to industry-level shocks and reflect local industrial specialization, while the shifts measure the size of those shocks. Growth rates r_{kgt} are gender-specific, given that aggregate demand shocks have been found to vary by gender in other settings (Olivetti and Petrongolo, 2014; Chauvin, 2024).

The first stage is

$$\Delta \ln e_{jgt} = \pi z_{jgt} + D'_{jgt} \tilde{\rho} + \tilde{\gamma}_t + u_{jgt}. \quad (4)$$

Identification of β requires that the shift-share instrument be correlated with local employment changes $\Delta \ln e_{jgt}$ (relevance) and uncorrelated with the local supply shifters in ε_{jgt}

(exclusion). Goldsmith-Pinkham, Sorkin, and Swift (2020) and Borusyak, Hull, and Jaravel (2022) show two distinct sufficient conditions for the IV identification assumption to hold: exogenous exposure shares s_{kjt0} for some set of industries, or a large number of as-good-as-random industry-level shocks r_{kgt} . We follow Borusyak, Hull, and Jaravel (2022) and rely on the latter. In that framework, the shift-share IV estimator is equivalent to a weighted IV regression at the industry-period level, in which the relevant unobservable is the exposure-weighted average of local supply shifters, $\bar{\varepsilon}_{kgt} = \sum_j s_{kjt0} \varepsilon_{jgt} / \sum_j s_{kjt0}$.

In this context, the exclusion assumption is that conditional on the controls and $\bar{\varepsilon}_{kgt}$, national industry employment growth r_{kgt} is as good as randomly assigned. In other words, industries that grew faster nationally are not systematically those to which workers became more or less willing to supply labor (weighing industries by local-labor-market exposure). Under this framework, a specific industry can receive demand and supply shocks at the same time (e.g., a new technology that raises an industry’s labor demand while also making employment more attractive), as long as this relation does not hold on average across industries and periods. Importantly, the identification assumption does not imply restrictions on the exposure shares, which can then be correlated with supply shifters at the local level. This is attractive in our setting, given that some important shifters of labor supply are unobserved in our data but distributed unevenly across local labor markets. Examples include conditional cash transfers, which decreased labor force participation and were targeted to poorer communities (Parker and Todd, 2017) and migration to the U.S., which tended to originate in locations with international migrant networks (Munshi, 2003).^{11,12}

¹¹The alternative interpretation, based on exogenous shares (Goldsmith-Pinkham, Sorkin, and Swift, 2020), amounts to a parallel-trends assumption at the local labor market level for each of the industries k that are used to construct the Bartik shock: in the absence of the industry shocks, local labor markets with larger shares in a given industry would have followed the same trends as those with smaller shares. This is difficult to defend in our setting. Industrial specialization in Mexico has a strong geographic pattern, with manufacturing concentrated in the north and center-north. At an aggregate level, the exogenous-shares assumption requires, for example, that northern and southern LLMs would have followed similar wage trends to the rest of the Mexican labor market in the absence of NAFTA. This strategy is most credible when a small number of “tailored” industries provide the identifying variation, as detailed in Borusyak, Hull, and Jaravel (2025). Our instrument instead pools shocks across all 3-digit industries, which fits the many-shocks interpretation more naturally.

¹²We show the diagnostics suggested in Goldsmith-Pinkham, Sorkin, and Swift (2020) for the “exogenous shares” approach in Appendix Tables A3 and A4, and Figures A2 and A3. Panel A of A3 shows that most sectors in all specifications (male and female, including and excluding agriculture) have positive Rotemberg weights, meaning that the IV estimator brings in their corresponding β_k estimates with positive weights. However, some important diagnostics suggest the set of sectors we use is unlikely to satisfy the “exogenous shares” assumption, especially for women. Sectors with Rotemberg weights below zero account for a substantial share of the total estimate for men and women, as shown in Panel D: negative-weight sectors contribute 27% and 16% of the overall estimate for women and men, respectively. For women, the β_k estimates using these weights tend to be negative, suggesting some of these estimates reflect variation from supply shocks rather than demand. Figure

The second requirement of the “many as-good-as-random shocks” identification framework is that Bartik shocks be constructed using many industry-years, with sufficiently dispersed exposure. Even with many industries in the sample, if one in particular accounts for a large share of employment in the average local labor market, it dominates the industry-level regression and the estimator effectively relies on few observations. We provide diagnostics to assess the validity of both identification assumptions below.

Mexico underwent an important structural transformation in the labor market over the study period, marked by a decreasing participation of agriculture in total employment (see Figure 1). Including agriculture in the sample of shocks raises concerns for both requirements above. First, some of the decline in agricultural employment reflects shifts in labor supply that are unobservable in our data. For instance, large-scale conditional cash transfers induced shifts away from labor market participation and into schooling, with the strongest effects among rural communities and women (Parker and Todd, 2017). Because these shifts concentrate in local labor markets specialized in agriculture, and national agricultural employment declined over the same period, the agricultural shock is likely correlated with $\bar{\varepsilon}_{kgt}$. Second, agriculture accounts for a large share of employment in the average local labor market, which lowers the effective number of shocks. For these reasons, in addition to our baseline estimates, we present results excluding agricultural employment from employment totals and from the set of sectors used to construct the shift-share instrument. When excluding agriculture, exposure shares are computed relative to non-agricultural employment.

Following Borusyak, Hull, and Jaravel (2022), we show several summary statistics on the distribution of shares to validate the set of sectors used in estimation. We first focus on the inverse Herfindahl index of the employment-weighted industry by period weights $s_{kt0} = \sum_j e_{jt} s_{kjt0}$, with e_{jt} the employment in LLM j at time t , which measures the effective number of shocks. Table 1 shows an effective sample size of 11 industry-periods with all sectors, and 65 when excluding agriculture, meaning that removing agriculture reduces the influence of a single shock and makes the “many uncorrelated shocks” assumption more likely to hold in the data.¹³ Consistent with this increased dispersion of exposure shares, removing agricul-

A2 shows that β_k estimates with positive Rotemberg weights tend to be similar to each other and close to the overall estimate for males; however, for females, Figure A3 shows important sectors with large negative weights even when removing agriculture. As mentioned above, supply and demand shocks appearing in the same sectors can be accounted for in the “many-shocks” approach, but call into question the identification assumption under “exogenous shares”.

¹³It may seem counterintuitive that excluding sectors makes the shock-level regression have a larger effective number of shocks. However, as Borusyak, Hull, and Jaravel (2022) explain, identification relies on a large number of shocks, while each shock affects a “small” number of local labor markets. This condition is better satisfied when excluding sectors that have relatively large shares in a large number of labor markets.

ture from the set of sectors lowers the size of the largest 3-digit industry-period share by an order of magnitude, from 0.16 to .016.

TABLE 1: SHOCK-LEVEL SAMPLE DIAGNOSTICS

	Male		Female	
	(1) All Sectors	(2) No Agriculture	(3) All Sectors	(4) No Agriculture
Descriptives: Shocks r_{kgt}				
Mean	.012	.615	.035	.765
Std. Dev.	.655	.529	.781	.600
IQR	.973	.604	1.251	.708
Descriptives: Shares s_{kt0}				
$1/HHI_{3D}$	11.628	65.492	11.628	65.492
$1/HHI_{1D}$	2.970	11.388	2.970	11.388
Largest share, 3-digit	.166	.016	.166	.016
Largest share, 1-digit	.566	.062	.566	.062
# shocks	264	246	264	246
# industries	88	82	88	82
# NAICS groups	23	22	23	22

Note: This table reports summary statistics for the shocks and shares used to construct Bartik instruments in our sample. A shock is a 3-digit NAICS industry-period pair. The mean, standard deviation, and interquartile range of national employment growth rates are weighted by average exposure shares, as in Borusyak, Hull, and Jaravel (2022). $1/HHI$ is the inverse Herfindahl-Hirschman index of s_{kt0} , computed across 3-digit industry-periods (3D) and 1-digit sector-periods (1D). The largest-share rows report the maximum s_{kt0} at each level of aggregation. Exposure shares are common across genders, so concentration statistics are identical in the male and female columns. Columns (2) and (4) exclude agriculture from employment totals, exposure shares, and the set of industries.

While the as-good-as-random assumption cannot be tested directly, it implies that the shocks should be uncorrelated with predetermined characteristics of the local labor markets exposed to them. We test this with shock-level balance regressions of exposure-weighted start-of-period characteristics on r_{kgt} . We first focus on balance for a set of variables that we can measure in our complete estimation sample: start-of-period wages and measures of social vulnerability. For men, none of these characteristics is significantly correlated with the shocks, as shown in Table 2. For women, Table 3 shows that the start-of-period share of individuals neither working nor studying (NEET) is correlated with the shocks. However, this correlation is substantially reduced when we exclude agriculture, and the coefficient is small. Crucially, shocks are unrelated to baseline levels of log wages, the outcome variable. Appendix tables

A1 and A2 expand the set of variables we use to test for balance by splitting the sample by year and including variables that we cannot reconstruct in every Census wave. Again, we find that shocks are unrelated to baseline wages, suggesting that the locations our estimation strategy is comparing had similar labor market conditions before the shocks. We also perform a pre-trend test, comparing employment trends in the first period (1990-2000) across locations with different shock levels in the next periods (2000-2010) and (2010-2020). Importantly, we find no relationship between growth rates in the first period and shocks in the second, pointing to unmodeled pre-trends not being a large threat to estimation. Some baseline social characteristics differ in some years across locations, but these correlations are very small in magnitude. This is the case for the share of single-parent households for the periods ending in 2010 and 2020, and the share of NEETs in 2020. When we look for correlations between shocks and skilled-by-gender bins, we find shocks tended to significantly slant towards locations with more skilled men and women in 2010, but the effects are quantitatively small given we control for education bins in the aggregate population.

TABLE 2: SHOCK BALANCE, MALES

	All Sectors		No Agriculture	
	Coeff	SE	Coeff	SE
Start-of-period % single-parent households	0.0001	(0.0002)	0.0001	(0.0002)
Start-of-period % NEETs	0.005	(0.004)	-0.003	(0.004)
Start-of-period log wage	0.002	(0.023)	-0.010	(0.018)
Observations	261		258	

Note: This table shows coefficients of regressions of industry-by-year employment growth rates on exposure-weighted characteristics of males at the start of the period corresponding to the shock, as in Borusyak, Hull, and Jaravel (2022). Each coefficient comes from a separate regression, at the industry-period level. Robust standard errors in parentheses. Controls include shares by age bin and education bin, the rural population share, and year-level fixed effects. Columns 3 and 4 exclude agriculture from employment totals and from the set of industries. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

We adjust inference to account for the shock-level variation that underlies the estimation, as suggested by Adão, Kolesár, and Morales (2019) and Borusyak, Hull, and Jaravel (2022). Borusyak, Hull, and Jaravel (2022) show that the shift-share IV estimator of (2) is numerically equivalent to an IV regression at the level of the shocks, but in the shock-level regression, instrumenting with r_{kgt} and weighting by s_{kt0} reproduces the shift-share IV estimate of β while delivering correct inference. These exposure-robust standard errors account for the correlation induced across local labor markets that share industrial specialization, and are our preferred measure of coefficient uncertainty.

TABLE 3: SHOCK BALANCE, FEMALES

	All Sectors		No Agriculture	
	Coeff	SE	Coeff	SE
Start-of-period % single-parent households	-0.001	(0.002)	-0.0001	(0.002)
Start-of-period % NEETs	0.051***	(0.017)	0.020*	(0.012)
Start-of-period log wage	0.018	(0.043)	-0.012	(0.023)
Observations	261		258	

Note: This table shows coefficients of regressions of industry-by-year employment growth rates on exposure-weighted characteristics of females at the start of the period corresponding to the shock, as in Borusyak, Hull, and Jaravel (2022). Each coefficient comes from a separate regression, at the industry-period level. Robust standard errors in parentheses. Controls include shares by age bin and education bin, the rural population share, and year-level fixed effects. Columns 3 and 4 exclude agriculture from employment totals and from the set of industries. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5 Results

We now show and discuss estimation results. We show first-stage estimates for both men and women, then second-stage estimates, and then turn to heterogeneity within women.

5.1 First Stage Estimates

Table 4 reports estimates of the first stage in Equation (4) for men and women, with and without controls, in the samples including and excluding agriculture. The coefficient measures the percent change in local employment associated with a 1 percentage point increase in the Bartik shock instrumental variable, the share-weighted average of national industry growth rates. In the sample including agriculture, a 1 percentage point increase in the Bartik shocks has statistically significant positive effects on employment, increasing it by 0.56% for males and 0.69% for females in the specifications with controls. While we cannot reject both estimates are equal, coefficients for females are slightly larger than for males when shifts from agriculture to other sectors are not counted as employment increases. This suggests that when faced with a similar shock, female entry into employment may be stronger than for males; we return to this in the discussion in Section 5.2. When excluding agriculture, coefficients are positive, smaller, and similar across genders, with a 0.49% response for men and 0.48% for women in response to a 1pp change in the Bartik shock. These estimates are consistent with the fact that Bartik shocks reflect positive shocks to local employment, as required by the IV strategy.

Controls affect estimates for women, with point estimates falling from 0.86 to 0.69. This pattern indicates that the LLM-level controls are correlated with Bartik shocks and independently predict female employment growth, i.e., controls are likely accounting for some labor supply confounders. We therefore treat the specifications with controls as our preferred ones. Exposure-robust F-statistics all clear the customary critical value of 10, suggesting sufficient instrument strength.

TABLE 4: FIRST STAGE ESTIMATES

	$\Delta \ln(emp)$			
	Male		Female	
	(1)	(2)	(3)	(4)
Shock level (With Agriculture)				
Employment Bartik	0.549*** (0.158)	0.557*** (0.154)	0.856*** (0.240)	0.691*** (0.129)
Observations	261	261	261	261
F-stat	12.086	13.069	12.710	28.874
Controls	No	Yes	No	Yes
Shock level (No Agriculture)				
Employment Bartik	0.356** (0.174)	0.492*** (0.087)	0.561 (0.355)	0.480*** (0.147)
Observations	258	258	258	258
F-stat	4.166	32.195	2.496	10.590
Controls	No	Yes	No	Yes

Note: This table reports first-stage estimates of Equation (4). The dependent variable is the 10-year change in log local labor market employment for each gender, and Employment Bartik is the shift-share instrument z_{jgt} in Equation (3). Estimates come from the equivalent shock-level regression of Borusyak, Hull, and Jaravel (2022), and observations are industry-period pairs. Exposure-robust standard errors in parentheses. F-stat is the exposure-robust first-stage F-statistic. Controls include start-of-period shares by age bin, education bin, and the rural population share. All columns include period fixed effects. The No Agriculture panel excludes agriculture from employment totals and from the set of industries used to construct the Bartik shock. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Inverse Supply Elasticity Estimates

Table 5 reports shift-share IV estimates of the inverse local labor supply elasticity β in Equation (2). In our preferred specification, an increase of 1% in male employment driven by local demand shocks is accompanied by a 0.36% increase in male wages, significant at the 10% level using exposure-robust standard errors. The 95% confidence interval, [-0.03, 0.76],

excludes values of β at or above one, which rules out inelastic male local labor supply in the sense of Figure 2. Our findings are then consistent with male local labor demand shocks being accommodated mostly through employment, with some evidence of wages also rising. In the U.S. setting, Adão, Kolesár, and Morales (2019) use similar techniques and data and arrive at an inverse elasticity of labor supply of approximately 0.8, i.e., also elastic but to a lesser extent than our estimated male labor supply.

TABLE 5: INVERSE LOCAL LABOR SUPPLY ELASTICITY ESTIMATES

	$\Delta \ln(wage)$			
	Male		Female	
	(1)	(2)	(3)	(4)
Shock level (With Agriculture)				
$\Delta \ln(emp)$	0.486 (0.335)	1.350*** (0.375)	-0.343*** (0.085)	-0.211 (0.168)
Observations	261	261	261	261
First-stage F-stat	12.087	13.069	12.710	28.874
Controls	No	Yes	No	Yes
Shock level (No Agriculture)				
$\Delta \ln(emp)$	0.636 (0.524)	0.361* (0.202)	-0.126 (0.242)	0.097 (0.244)
Observations	258	258	258	258
First-stage F-stat	4.166	32.195	2.496	10.590
Controls	No	Yes	No	Yes

Note: This table reports shift-share IV estimates of the inverse local labor supply elasticity β in Equation (2). The dependent variable is the 10-year change in the log of average monthly labor income in the local labor market for each gender, in constant 2015 pesos and excluding the top and bottom 1% of the wage distribution within each local labor market-year. $\Delta \ln(emp)$ is the 10-year change in log employment, instrumented with z_{jgt} . Estimates come from the equivalent shock-level regression of Borusyak, Hull, and Jaravel (2022). Observations are industry-period pairs. Exposure-robust standard errors in parentheses. The first-stage F-statistic is from Table 4. Controls include start-of-period shares by age bin, education bin, and the rural population share. All columns include period fixed effects. The No Agriculture panel excludes agriculture from employment totals and from the set of industries used to construct the Bartik shock. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

While we prefer estimates that exclude agriculture, the contrast between specifications with and without the sector allows us to speculate about the role of agricultural labor in responding to local demand shocks. Estimates with agriculture yield a much larger male estimate: $\beta = 1.35$, which implies an elasticity of about 0.74 and inelastic local supply. Given the decrease in agriculture's share of total employment, reallocation of employment out of agriculture is a possible driver of this difference. If non-agricultural demand shocks draw workers from

lower-paid agricultural employment to higher-wage work in other sectors, our model would interpret that as a source of inelasticity when agricultural employment is included in local totals, since total employment would remain constant while average wages rise. Our data do not allow us to observe individual workers moving across sectors from one Census wave to the next, so directly testing this mechanism remains outside the scope of this paper.¹⁴

For women, point estimates are close to zero and statistically insignificant: in our preferred specification, $\beta = 0.097$. The confidence intervals are consistent with highly elastic female local labor supply, meaning that local demand shocks raise female employment with little or no measurable wage response. These results align with evidence that female employment responds more strongly than male employment to demand shocks in Mexican manufacturing (Trachtenberg, 2019; Chen, Novy, and Solórzano, 2025), and with Chauvin (2024), who finds that women in Brazil adjust to local shocks mainly along the participation margin.

Taken together, our results then point to a non-agricultural labor supply in Mexico that is relatively elastic. As the descriptive exercises in Aldeco et al. (2024) show, real wage distributions across Mexican local labor markets have remained quite stable since the 1990s. While this might seem puzzling in the face of several reforms and external shocks that have shifted labor demand in the country (NAFTA, the Global Financial Crisis, China’s entry into the WTO, the U.S.-China trade war), our estimates suggest an elastic supply of labor as a contributing factor to these apparent weak wage responses. Our results also help explain the general pattern in the literature, by which trade shocks in Mexican manufacturing usually induce larger employment than wage responses. The apparent larger elasticity of female supply suggests that the pool of non-participating women is a particularly important margin of adjustment, which we explore next by marital status and the presence of young children.

5.3 Heterogeneity by Marriage Status and Children

Table 6 reports first-stage and inverse supply elasticity estimates for women, split by marital status and by the presence of children younger than 6 in the household. Local demand shocks have positive and statistically significant effects on employment in all four groups. The main contrast is between married and non-married women: within each childcare-demand bin, the first-stage coefficient is larger for married women. A 1pp increase in the Bartik shock in-

¹⁴Sectoral reallocation is a relevant margin by which local labor markets accommodate sector-biased external shocks, as shown in Autor et al. (2025). In a recent episode in our setting, sectoral reallocation within local labor markets contributed around 20% to the growth in manufacturing employment observed between 2020 and 2022 in the northern region, in response to surging demand from the U.S.; see Banco de México (2022).

creases employment by 0.70% for married women with young children, compared to 0.49% for non-married women with young children, and by 0.61% for married women without young children, compared to 0.22% for their non-married counterparts. The first stage for non-married women without young children is weaker, with an F-statistic of 3.1 and a coefficient significant only at the 10% level. Larger local employment responses among married women are consistent with them constituting a larger pool of workers available to local employers. Married women’s labor supply has traditionally been found to be highly responsive to labor market conditions, partly reflecting their role as secondary earners in the household (Killingsworth and Heckman, 1986; Blau and Kahn, 2007), and lower-earning spouses are more likely to move in and out of employment than other workers (McClelland, Mok, and Pierce, 2014). This pattern also aligns with Chauvin (2024), who finds that labor force participation is a relatively important margin of adjustment to female labor demand shocks.

Inverse supply elasticity estimates are small and statistically insignificant for all four groups, ranging from 0.01 for married women with young children to 0.50 for non-married women without young children, in line with the estimate of 0.10 for all women in Table 5. Since our outcome is the average wage in the local labor market, a flat wage response can partly reflect the composition of new entrants. If local demand shocks draw into employment women whose wages are lower than the average for their group, average wages may remain constant even if the wages of already-employed women increase. Taken together, these results suggest that when a female labor demand shock hits a local labor market, married women contribute substantially to accommodating it through increased employment. The lack of wage responses across groups is harder to interpret, given that first-stage F-statistics fall below the customary critical value of 10 in three of the four subsamples. However, the general picture is consistent with our main results, i.e., aggregate female local labor supply looks elastic.

5.4 Robustness

We perform a small set of additional exercises to assess robustness in Table 7. Monte, Redding, and Rossi-Hansberg (2018) show that local employment elasticities to demand shocks depend on the degree of connectedness of a local labor market to surrounding markets. To account for this dependence, columns (1) and (2) control for the share of commuters in each local labor market in each period. While our definition of commuting zones accounts for a large share of total commuting, LLMs can still display heterogeneity in commuter shares in employment that can, in principle, influence the estimates. Importantly, however, our data do

TABLE 6: INVERSE FEMALE LOCAL LABOR SUPPLY ELASTICITY ESTIMATES: HETEROGENEITY

	Female			
	Married		Single/Other	
	(1) children 0-5 years	(2) no children or 6+ years	(3) children 0-5 years	(4) no children or 6+ years
First Stage Estimates				
Employment Bartik	0.697*** (0.218)	0.611*** (0.210)	0.489*** (0.170)	0.224* (0.128)
Observations	258	258	258	258
F-stat	10.238	8.496	8.263	3.100
Controls	Yes	Yes	Yes	Yes
Inverse Local Labor Supply Elasticity Estimates				
$\Delta \ln(emp)$	0.012 (0.218)	0.171 (0.256)	0.170 (0.308)	0.503 (0.652)
Observations	258	258	258	258
Controls	Yes	Yes	Yes	Yes

Note: This table reports first-stage and shift-share IV estimates of the female local labor supply elasticity, separated by marital status and by the presence of children aged 0–5 in the household. Married includes married and cohabiting women; Single/Other includes single, widowed, and separated women. Columns (1) and (3) include women in households with children aged 0–5, and columns (2) and (4) include women in households without children or with children aged 6 or older. The top panel reports estimates of Equation (4), where the dependent variable is the 10-year change in log local labor market employment for each group and Employment Bartik is the shift-share instrument z_{jgt} in Equation (3). The bottom panel reports estimates of the inverse local labor supply elasticity β in Equation (2), where the dependent variable is the 10-year change in the log of average monthly labor income for each group, and $\Delta \ln(emp)$ is instrumented with z_{jgt} . Estimates come from the equivalent shock-level regression of Borusyak, Hull, and Jaravel (2022). Observations are industry-period pairs. Exposure-robust standard errors in parentheses. F-stat is the exposure-robust first-stage F-statistic. Controls include start-of-period shares by age bin and education bin and the rural population share. All columns include period fixed effects and exclude agriculture from employment totals and the set of industries used to construct the Bartik shock. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

not allow us to construct commuting flows across LLMs for the 1990-2000 period, which is an important source of local labor demand shocks (in particular due to the start of NAFTA). As a consequence, the instrument is weaker in this smaller sample, with lower first-stage F-statistics than in our baseline results. Despite the weaker first stages, the qualitative results remain the same when we include the commuting control. For men, when including agriculture, the inverse elasticity remains positive and significant at the 5% level, at 2.50, consistent

with a relatively inelastic male local labor supply. The significant elastic estimate when excluding agriculture also remains when including the additional control. For women, we again find no evidence of wage effects from demand shocks, although the smaller sample results in very imprecise estimates.

Last, columns (3) and (4) of Table 7 exclude some sectors that cannot be cleanly matched across sectoral employment catalogs throughout the study period, which we call “unmatched sectors”.¹⁵ Again, while first-stage F-statistics do fall, excluding the unmatched sector changes our estimates little: β is 1.48 for men with agriculture, compared to 1.35 in the corresponding specification in Table 5, and 0.5 when excluding it, compared to 0.36 in the baseline. Both of these estimates are statistically significant at the 5% level. Estimates for females also remain similar to baseline, with no significant wage effects. These results suggest that exposure to the unmatched sector captures real changes in local employment, likely in informal and service activities that Census respondents describe less precisely. The unmatched sector then contributes to the strength of our instrument but does not drive our estimates.

¹⁵The 1990 Census uses a classification that predates NAICS, with residual codes for insufficiently specified descriptions at most levels of aggregation. The 2000 Census uses a version of SCIAN with residual codes in only nine sectors. The 2010 and 2020 Censuses, based on SCIAN 2007 and SCIAN 2018, add residual codes for mining, information, educational and health services, as well as specific codes for street vendors and other itinerant workers. Under the exogenous-shares assumption, this sector has the largest Rotemberg weight when estimating the elasticity of males without agriculture, as seen in A4, highlighting the need to test for sensitivity to removing them.

TABLE 7: INVERSE LOCAL LABOR SUPPLY ELASTICITY ESTIMATES: ROBUSTNESS

	$\Delta \ln(wage)$			
	With Commuter Share Controls		Without unmatched sectors	
	(1)	(2)	(3)	(4)
Panel A: Men, with agriculture				
$\Delta \ln(emp)$	0.643 (0.793)	2.499** (1.108)	0.533 (0.404)	1.483*** (0.467)
Observations	174	174	258	258
First Stage F-stat	5.940	3.381	8.204	10.952
Controls	No	Yes	No	Yes
Panel B: Men, without agriculture				
$\Delta \ln(emp)$	1.420 (2.666)	0.803* (0.456)	1.082 (0.990)	0.509** (0.201)
Observations	172	172	255	255
First Stage F-stat	0.240	5.894	1.656	16.925
Controls	No	Yes	No	Yes
Panel C: Women, with agriculture				
$\Delta \ln(emp)$	-0.431 (0.448)	0.101 (1.082)	-0.318*** (0.081)	-0.205 (0.162)
Observations	174	174	258	258
First Stage F-stat	2.842	1.729	8.789	20.316
Controls	No	Yes	No	Yes
Panel D: Women, without agriculture				
$\Delta \ln(emp)$	1.329 (3.115)	0.723 (0.653)	-0.133 (0.219)	0.209 (0.370)
Observations	172	172	255	255
First Stage F-stat	0.284	2.724	1.632	4.288
Controls	No	Yes	No	Yes

Note: This table reports robustness checks for the shift-share IV estimates of the inverse local labor supply elasticity β in Equation (2). The dependent variable is the 10-year change in the log of average monthly labor income in the local labor market for each gender, in constant 2015 pesos and excluding the top and bottom 1% of the wage distribution within each local labor market-year. $\Delta \ln(emp)$ is the 10-year change in log employment, instrumented with z_{jgt} . Estimates come from the equivalent shock-level regression of Borusyak, Hull, and Jaravel (2022). Observations are industry-period pairs. Columns (1) and (2) add controls for commuter shares. Columns (3) and (4) exclude unmatched sectors (NAICS 999). Panel B excludes agriculture (NAICS 111) from employment totals, exposure shares, and the set of industries in all columns, following our preferred specification for women. Exposure-robust standard errors in parentheses. Controls include start-of-period shares by age bin and education bin and the rural population share. All columns include period fixed effects. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

6 Conclusion

This paper estimates the inverse elasticity of local labor supply in Mexico over 1990–2020. We combine harmonized Census microdata with local labor market definitions built from commuting flows, using the same method as U.S. commuting zones. We instrument local employment growth with gender-specific shift-share measures of labor demand, relying on many as-good-as-random industry shocks for identification and on exposure-robust inference. Our results point to an elastic non-agricultural local labor supply. For men, a 1% increase in employment driven by local demand shocks raises average wages by 0.36%, and we can rule out inelastic supply. This response is more elastic than comparable estimates for the U.S. For women, demand shocks raise employment, particularly among married women, with no detectable response of average wages. When agricultural employment is included, male supply appears inelastic. This is consistent with demand shocks drawing workers out of lower-paid agricultural jobs, although we cannot observe this reallocation directly.

These findings help reconcile two patterns in the Mexican evidence. Real wage distributions across local labor markets have been stable since 1990 despite large external shocks, and trade shocks tend to move employment more than wages. Our estimates suggest that local demand shocks and demand-side policies are more effective at creating jobs than at raising average wages. They also suggest that women outside employment are an important margin of adjustment for local labor markets. Raising average wages may then depend more on policies that shift labor supply and productivity, such as investments in human capital.

Several limitations point to directions for future work. First, our elasticity is local and reduced-form: it does not distinguish among different channels through which employment may change, i.e., migration, commuting, changes in participation, and sectoral reallocation within the local labor market. As Redding and Turner (2015) highlight, local changes in population and employment can reflect aggregate growth or spatial reallocation. In the same direction, Autor et al. (2025) show that positive measured shocks to local labor markets can mask different outcomes for incumbents and for mobile workers who move to take advantage of spatial differences in opportunities. Separating these channels is necessary to understand the distributional effects of local shocks. Second, because our outcome is the average local wage, flat wage responses may partly reflect the composition of new entrants rather than unchanged wages for incumbents. Worker-level panel data would help distinguish the two. Third, our estimates capture only the extensive margin and are silent about adjustments in hours and between formal and informal employment. These margins are likely important for

how Mexican local labor markets absorb demand shocks.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Claude Opus 5.5 and Grammarly for Edge version 14.1153.0 to assist with literature review, check spelling, improve clarity, and format BibTeX entries. After using these tools, the authors reviewed and edited the content as needed. They take full responsibility for the content of the publication.

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A Additional Tables

TABLE A1: BALANCE TESTS — MALE

Male	With Agriculture			No Agriculture		
	2000	2010	2020	2000	2010	2020
Start-of-period informality rate		0.024 (0.014) [0.084]	-0.036 (0.034) [0.288]		0.029 (0.017) [0.087]	0.009 (0.016) [0.606]
Employment growth 2000s		-0.007 (0.016) [0.685]	0.064 (0.024) [0.009]		-0.011 (0.017) [0.547]	0.050 (0.015) [0.001]
Start-of-period % households below extreme poverty line	-0.002 (0.014) [0.865]		-0.023 (0.012) [0.051]	0.006 (0.014) [0.669]		-0.008 (0.008) [0.329]
Start-of-period % households below poverty line	0.003 (0.011) [0.766]		-0.014 (0.012) [0.226]	0.008 (0.012) [0.528]		-0.020 (0.013) [0.131]
Start-of-period log wage	-0.016 (0.036) [0.653]	-0.016 (0.028) [0.579]	0.048 (0.048) [0.322]	-0.004 (0.025) [0.864]	-0.010 (0.028) [0.707]	-0.003 (0.021) [0.891]
Start-of-period % single-parent households	-0.0003 (0.0003) [0.408]	0.0003 (0.0002) [0.066]	0.001 (0.0003) [0.023]	-0.0001 (0.0004) [0.719]	0.0004 (0.0002) [0.065]	0.001 (0.0002) [0.001]
Start-of-period % NEETs	0.003 (0.005) [0.569]	-0.006 (0.004) [0.111]	0.008 (0.004) [0.027]	0.003 (0.005) [0.521]	-0.014 (0.004) [0.000]	0.002 (0.002) [0.396]
Start-of-period % skilled men	0.001 (0.001) [0.677]	0.002 (0.001) [0.028]	-0.003 (0.001) [0.033]	0.001 (0.001) [0.557]	0.003 (0.001) [0.023]	-0.003 (0.001) [0.031]
Start-of-period % employment among women	0.022 (0.008) [0.006]	0.0001 (0.007) [0.987]	0.022 (0.010) [0.022]	0.024 (0.010) [0.008]	-0.005 (0.010) [0.591]	0.003 (0.010) [0.769]

Note: This table shows the coefficients of regressions of industry-by-year employment growth rates on exposure-weighted characteristics of males at the start of the period corresponding to the shock, as in Borusyak, Hull, and Jaravel (2022), estimated separately for each period. Column headers indicate the end year of each 10-year period. Each coefficient comes from a separate regression at the industry level. “Employment growth 2000s” is the 1990–2000 change in log male employment. Empty cells correspond to variables that cannot be constructed from the Census wave at the start of the period. Robust standard errors in parentheses and p-values in brackets. Controls include shares by age bin and education bin and the rural population share. The No Agriculture columns exclude agriculture from employment totals and the set of industries.

TABLE A2: BALANCE TESTS — FEMALE

Female	With Agriculture			No Agriculture		
	2000	2010	2020	2000	2010	2020
Start-of-period informality rate		0.009 (0.043) [0.842]	-0.009 (0.012) [0.479]		0.027 (0.031) [0.382]	-0.011 (0.014) [0.466]
Employment growth 2000s		-0.282 (0.332) [0.396]	0.060 (0.079) [0.445]		-0.082 (0.161) [0.608]	0.103 (0.080) [0.199]
Start-of-period % households below extreme poverty line	0.033 (0.014) [0.016]		-0.005 (0.006) [0.389]	0.017 (0.013) [0.184]		-0.006 (0.007) [0.356]
Start-of-period % households below poverty line	0.017 (0.012) [0.162]		-0.010 (0.008) [0.210]	0.019 (0.012) [0.126]		-0.013 (0.011) [0.221]
Start-of-period log wage	-0.018 (0.020) [0.374]	0.005 (0.050) [0.920]	0.008 (0.013) [0.525]	-0.014 (0.026) [0.604]	-0.027 (0.022) [0.225]	0.012 (0.016) [0.459]
Start-of-period % single-parent households	0.002 (0.003) [0.563]	0.002 (0.003) [0.378]	0.001 (0.002) [0.664]	-0.001 (0.002) [0.444]	0.005 (0.003) [0.072]	0.001 (0.002) [0.737]
Start-of-period % NEETs	0.023 (0.017) [0.186]	-0.007 (0.009) [0.397]	0.005 (0.007) [0.496]	0.005 (0.008) [0.557]	-0.007 (0.011) [0.535]	0.009 (0.008) [0.277]
Start-of-period % skilled women	-0.001 (0.001) [0.262]	0.004 (0.002) [0.032]	-0.002 (0.002) [0.198]	-0.001 (0.001) [0.392]	0.004 (0.002) [0.071]	-0.002 (0.002) [0.255]
Start-of-period % employment among women	-0.002 (0.012) [0.852]	0.027 (0.012) [0.022]	-0.002 (0.007) [0.798]	0.006 (0.010) [0.541]	0.017 (0.011) [0.145]	-0.005 (0.009) [0.550]

Note: This table shows the coefficients of regressions of industry-by-year employment growth rates on exposure-weighted characteristics of females at the start of the period corresponding to the shock, as in Borusyak, Hull, and Jaravel (2022), estimated separately for each period. Column headers indicate the end year of each 10-year period. Each coefficient comes from a separate regression at the industry level. “Employment growth 2000s” is the 1990–2000 change in log female employment. Empty cells correspond to variables that cannot be constructed from the Census wave at the start of the period. Robust standard errors in parentheses and p-values in brackets. Controls include shares by age bin and education bin and the rural population share. The No Agriculture columns exclude agriculture from employment totals and the set of industries.

TABLE A3: ROTEMBERG WEIGHTS AND RELATED STATISTICS

	Female						Male					
	With Agriculture			No Agriculture			With Agriculture			No Agriculture		
	(1)			(2)			(3)			(4)		
Panel A: Negative and positive weights												
	Sum	Mean	Share	Sum	Mean	Share	Sum	Mean	Share	Sum	Mean	Share
Negative	-0.375	-0.009	0.214	-0.737	-0.018	0.298	-0.429	-0.010	0.231	-0.303	-0.012	0.189
Positive	1.375	0.030	0.786	1.737	0.039	0.702	1.429	0.032	0.769	1.303	0.021	0.811
Panel B: Correlations of Industry Aggregates												
Corr($\hat{\alpha}_k, \mathbf{r}_k$)	-0.013			-0.018			0.009			0.088		
Corr($\hat{\alpha}_k, \hat{\beta}_k$)	0.001			0.037			0.034			-0.020		
Corr($\hat{\alpha}_k, \hat{F}_k$)	0.517			0.168			0.206			0.023		
Corr($\hat{\alpha}_k, \text{Var}(z_k)$)	0.936			0.412			0.404			0.421		
Corr($\mathbf{r}_k, \hat{\beta}_k$)	-0.062			-0.413			0.859			0.200		
Corr($\mathbf{r}_k, \hat{F}_k$)	-0.073			-0.037			0.058			0.033		
Corr($\mathbf{r}_k, \text{Var}(z_k)$)	0.010			-0.060			-0.025			-0.072		
Corr($\hat{\beta}_k, \hat{F}_k$)	0.004			0.085			0.036			0.257		
Corr($\hat{\beta}_k, \text{Var}(z_k)$)	-0.001			0.014			0.001			0.030		
Corr($\hat{F}_k, \text{Var}(z_k)$)	0.502			0.016			0.177			0.218		
Panel C: Variation across years in α_k												
	Sum	Mean		Sum	Mean		Sum	Mean		Sum	Mean	
2000	0.592	0.006		0.573	0.006		0.678	0.007		0.176	0.002	
2010	0.405	0.004		0.425	0.004		-0.072	-0.001		-0.143	-0.001	
2020	0.002	0.000		0.002	0.000		0.394	0.004		0.967	0.009	
Panel D: Estimates of β_k for positive and negative weights												
	α -Sum	β -Share	Mean	α -Sum	β -Share	Mean	α -Sum	β -Share	Mean	α -Sum	β -Share	Mean
Negative	0.222	-1.054	-1.394	0.055	0.268	-2.410	-0.088	-0.065	-0.983	-0.003	-0.163	0.020
Positive	-0.433	2.054	0.764	0.152	0.732	-0.212	1.434	1.065	0.618	0.024	1.163	0.029

Note: This table shows diagnostics for Rotemberg weights and their corresponding estimates β_k , for the set of all sectors and excluding agriculture, and by gender. Panel A shows summary statistics of Rotemberg weights, by positive- and negative-weighted sectors. Panel B shows industry-level correlations between Rotemberg weights ($\hat{\alpha}_k$), industry growth rates (r_k), estimates of β_k , and the variation of sector k 's shares in the distribution of local labor markets $\text{Var}(z_k)$. Panel C shows summary statistics of α_k by year. Panel D shows summary statistics of β_k estimates by Rotemberg weight sign. α -Sum is the $\hat{\alpha}_k$ weighted sum of β_k ; β -Share is the contribution to the β estimate, and the mean column refers to the average β_k value.

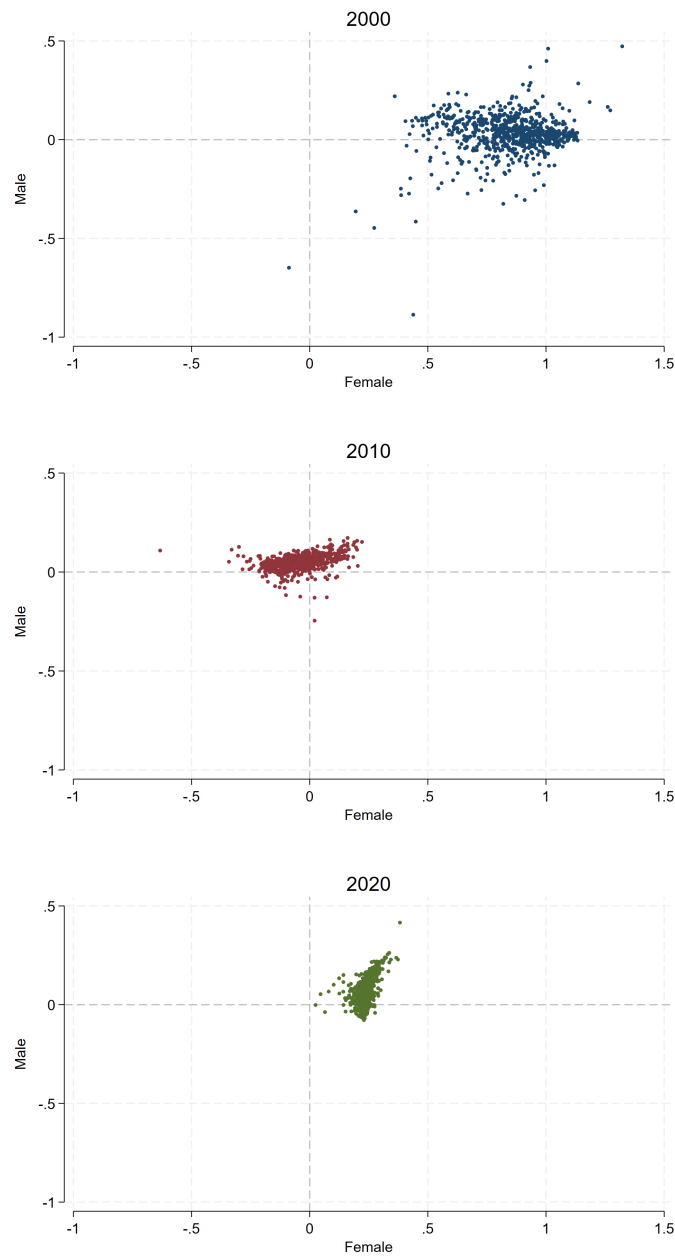
TABLE A4: ROTEMBERG WEIGHTS AND RELATED STATISTICS (CONTINUATION: PANEL D)

Panel E: Top 5 Rotemberg weight industries					
	$\hat{\alpha}_k$	r_k	$\hat{\beta}_k$	95% CI	Ind Share
(1) Female — With Agriculture					
Agriculture	1.012	0.834	-0.446	(-0.60, -0.30)	20.296
Retail and street trade of household goods [...]	0.074	-0.590	-0.269	(-0.40, -0.20)	3.892
Personal services, parking and lodging services for motor vehicles [...]	0.056	1.832	-0.338	(-0.50, -0.20)	0.784
Husbandry and exploitation of animals and aquaculture	0.036	1.891	-0.432	(-2.70, 0.30)	1.766
Drilling of oil and gas wells, and services related to mining	0.033	-0.963	1.190	(0.50, 2.10)	0.258
(2) Female — No Agriculture					
Husbandry and exploitation of animals and aquaculture	0.404	1.943	-0.386	(-0.90, 0.00)	2.899
Residential, non-residential construction, and residential self-construction	0.315	1.362	0.061	(-0.30, 0.40)	7.723
Retail and street trade of household goods, computers, decorative items ...	0.160	-0.640	-0.009	(-0.20, 0.20)	4.864
Insufficiently specified descriptions of the activity subsector of sector 11	0.146	-3.054	-0.238	(-9.10, 0.40)	1.015
Drilling of oil and gas wells, and services related to mining	0.095	-1.040	1.940	(0.80, 7.40)	0.332
(3) Male — With Agriculture					
Insufficiently specified descriptions of the activity subsector of sector 31-33	0.196	0.029	1.525	(1.00, 2.70)	2.195
Insufficiently specified descriptions of the activity subsector of sector 11	0.145	-3.700	2.066	(1.40, 3.80)	0.563
Agriculture	0.141	-0.031	1.697	(1.20, 2.70)	20.425
Drilling of oil and gas wells, and services related to mining	0.107	-1.727	2.164	(1.40, 4.60)	0.258
Manufacturing of transportation equipment and motor vehicle parts	0.099	0.720	0.648	(0.00, 1.70)	0.917
(4) Male — No Agriculture					
Unspecified economic activity	0.182	0.662	0.002	N/A	4.782
Insufficiently specified descriptions of the activity subsector of sector 31-33	0.146	0.627	0.043	(1.10, 5.60)	2.505
Retail and mobile trade of groceries, food, beverages, ice, and tobacco	0.084	0.661	-0.003	N/A	3.479
Residential, non-residential construction, and residential self-construction	0.073	0.353	-0.008	N/A	7.330
Formal food and beverage preparation services and; nightclubs	0.072	0.503	-0.002	N/A	3.176

Note: This table continues Table A3. Panel E reports the five industries with the largest Rotemberg weights in each specification, following Goldsmith-Pinkham, Sorkin, and Swift (2020). $\hat{\alpha}_k$ is the Rotemberg weight, r_k is the national employment growth rate of industry k , $\hat{\beta}_k$ is the just-identified IV estimate of β using industry k 's exposure share as the only instrument, and 95% CI is its confidence interval. Specifications (2) and (4) exclude agriculture from employment totals, exposure shares, and the set of industries.

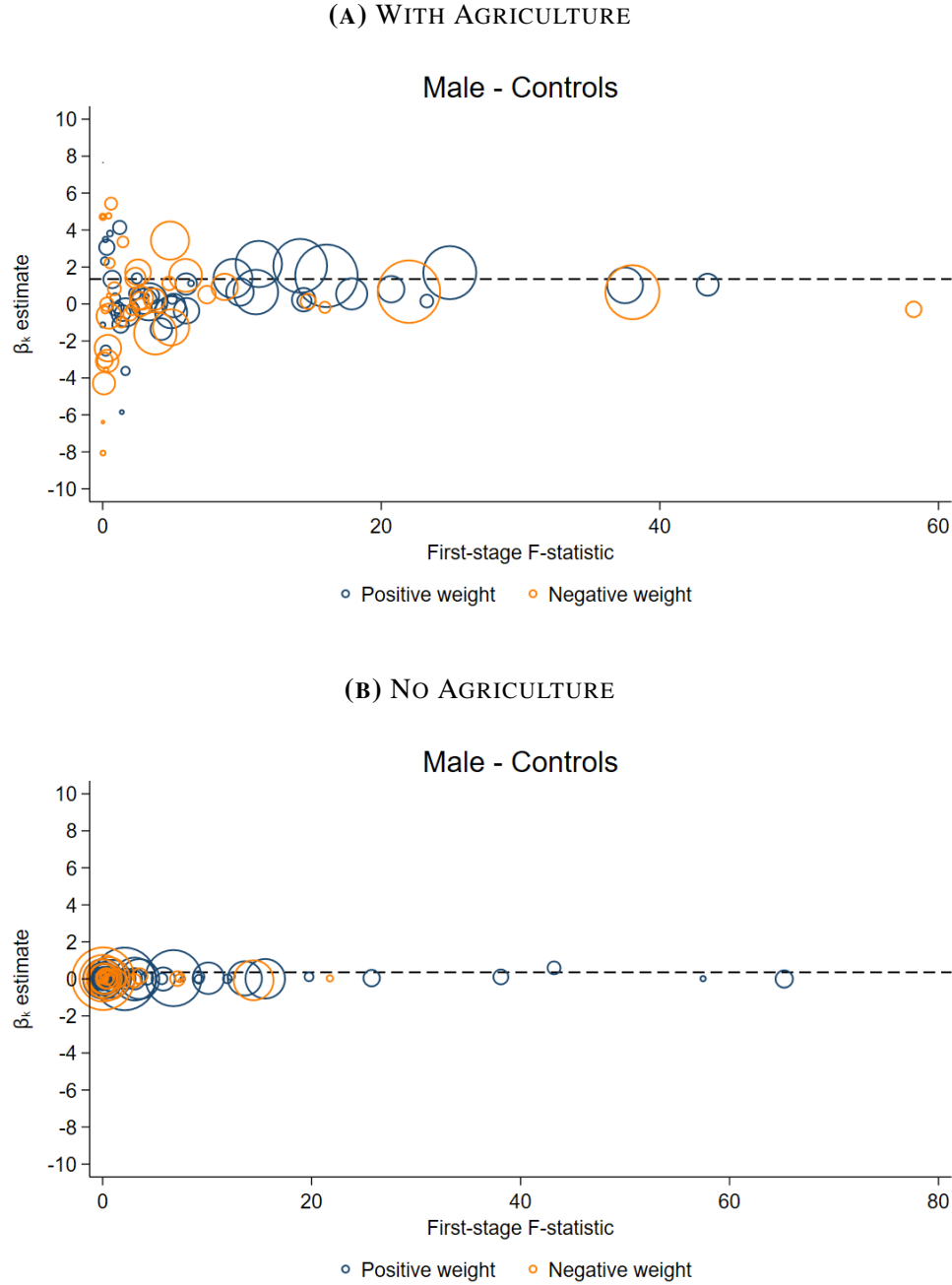
B Additional Figures

FIGURE A1: MALE AND FEMALE BARTIK SCATTERPLOTS



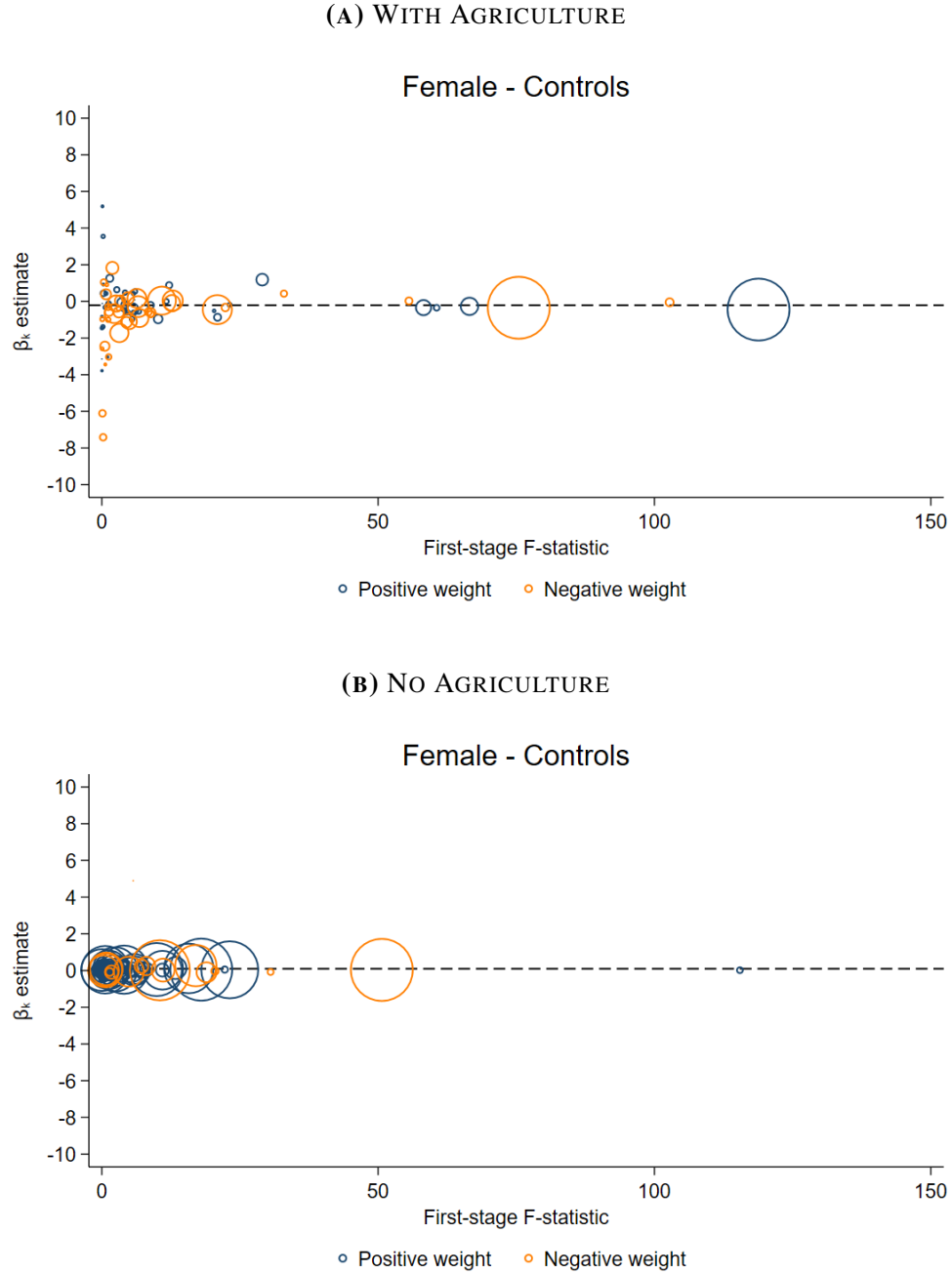
Note: This figure plots the male (vertical axis) against the female (horizontal axis) shift-share instrument z_{jgt} from Equation (3) for each local labor market, separately for each 10-year period, labeled by its end year. Instruments include all sectors.

FIGURE A2: HETEROGENEITY IN BETA ESTIMATES AND ROTEMBERG WEIGHTS — MALE



Note: This figure plots, for each industry k , the just-identified IV estimate $\hat{\beta}_k$ that uses industry k 's exposure share as the only instrument against its first-stage F-statistic, for men, following Goldsmith-Pinkham, Sorkin, and Swift (2020). Marker size is proportional to the absolute value of the industry's Rotemberg weight $\hat{\alpha}_k$; blue markers indicate positive weights and red markers negative weights. The dashed line shows the corresponding shift-share IV estimate of β in Table 5. All estimates include start-of-period shares by age bin and education bin, the rural population share, and period fixed effects. Panel B excludes agriculture from employment totals and the set of industries.

FIGURE A3: HETEROGENEITY IN BETA ESTIMATES AND ROTEMBERG WEIGHTS — FEMALE



Note: This figure plots, for each industry k , the just-identified IV estimate $\hat{\beta}_k$ that uses industry k 's exposure share as the only instrument against its first-stage F-statistic, for women, following Goldsmith-Pinkham, Sorkin, and Swift (2020). Marker size is proportional to the absolute value of the industry's Rotemberg weight $\hat{\alpha}_k$; blue markers indicate positive weights and red markers negative weights. The dashed line shows the corresponding shift-share IV estimate of β in Table 5. All estimates include start-of-period shares by age bin and education bin, the rural population share, and period fixed effects. Panel B excludes agriculture from employment totals and the set of industries.